US012416759B2

(12) United States Patent
Meyer Timmerman Thijssen et al.

(10) Patent No.: US 12,416,759 B2
(45) Date of Patent: Sep. 16, 2025

(54) **METHOD FOR FORMING MULTI-DEPTH OPTICAL DEVICES**

(71) Applicant: **Applied Materials, Inc.**, Santa Clara, CA (US)

(72) Inventors: **Rutger Meyer Timmerman Thijssen**, Sunnyvale, CA (US); **Guannan Chen**, Belmont, CA (US)

(73) Assignee: **Applied Materials, Inc.**, Santa Clara, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 303 days.

(21) Appl. No.: **18/076,242**

(22) Filed: **Dec. 6, 2022**

(65) **Prior Publication Data**

US 2023/0194784 A1 Jun. 22, 2023

Related U.S. Application Data

(60) Provisional application No. 63/265,910, filed on Dec. 22, 2021.

(51) **Int. Cl.**

G02B 1/00	(2006.01)
F21V 8/00	(2006.01)
G02B 5/18	(2006.01)
G02B 6/122	(2006.01)
G02B 6/136	(2006.01)
G02B 6/42	(2006.01)
G02B 27/00	(2006.01)

(Continued)

(52) **U.S. Cl.**

CPC ***G02B 6/122*** (2013.01); ***G02B 1/002*** (2013.01); ***G02B 5/1814*** (2013.01); ***G02B 5/1866*** (2013.01); ***G02B 6/0016*** (2013.01); ***G02B 6/136*** (2013.01); ***G02B 6/4204*** (2013.01); ***G02B 27/0081*** (2013.01); *G02B 6/124* (2013.01); *G02B 27/0172* (2013.01)

(58) **Field of Classification Search**

CPC G02B 1/002; G02B 5/1814; G02B 5/1857; G02B 5/1866; G02B 6/0016; G02B 6/0065; G02B 6/122; G02B 6/124; G02B 6/136; G02B 6/4204; G02B 27/0081; G02B 27/0172; G02B 2027/0194

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

8,506,829 B2 * 8/2013 Zhou B82Y 20/00 216/2
9,874,700 B2 * 1/2018 Sugama G02B 6/124

(Continued)

OTHER PUBLICATIONS

International Search Report and Written Opinion dated Apr. 24, 2023 for PCT application No. PCT/US2022/080977.

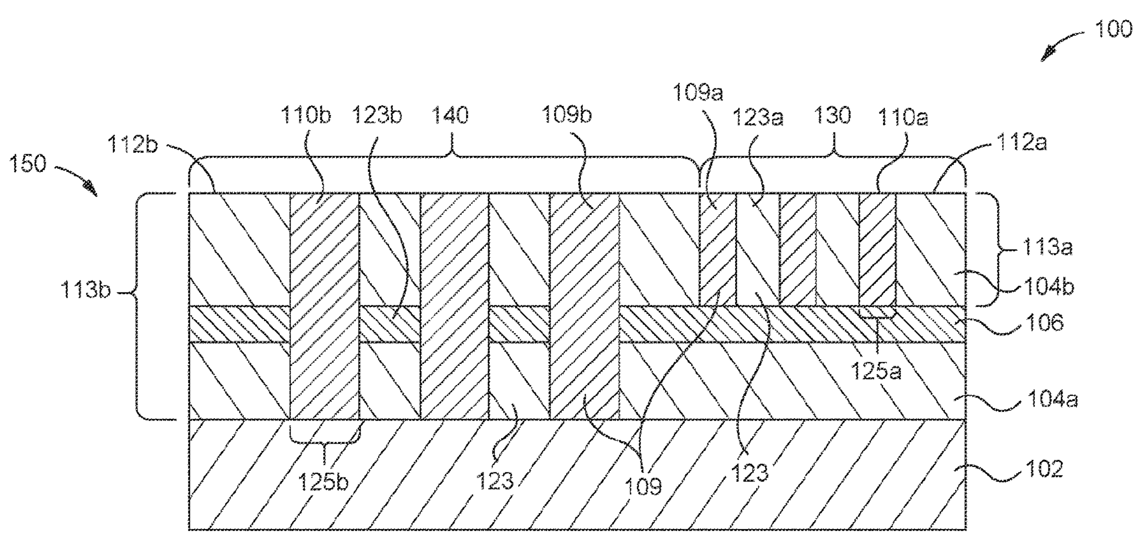
Primary Examiner — Ryan A Lepisto

(74) *Attorney, Agent, or Firm* — Patterson + Sheridan, LLP

(57) **ABSTRACT**

An optical device and a method of forming an optical device having multi-depth optical device structures with a refractive index greater than or equal to 2.0 are provided. The optical devices and method include forming first matrix stack structures and second matrix stack structures. Adjacent first matrix stack structures form first vias having a first depth and adjacent second matrix stack structures form second vias having a second depth. The first depth of the first vias being different than the second depth of the second vias provides from the formation of submicron, multi-depth optical device structures when the vias are backfilled with a device material.

20 Claims, 7 Drawing Sheets



(51) **Int. Cl.**

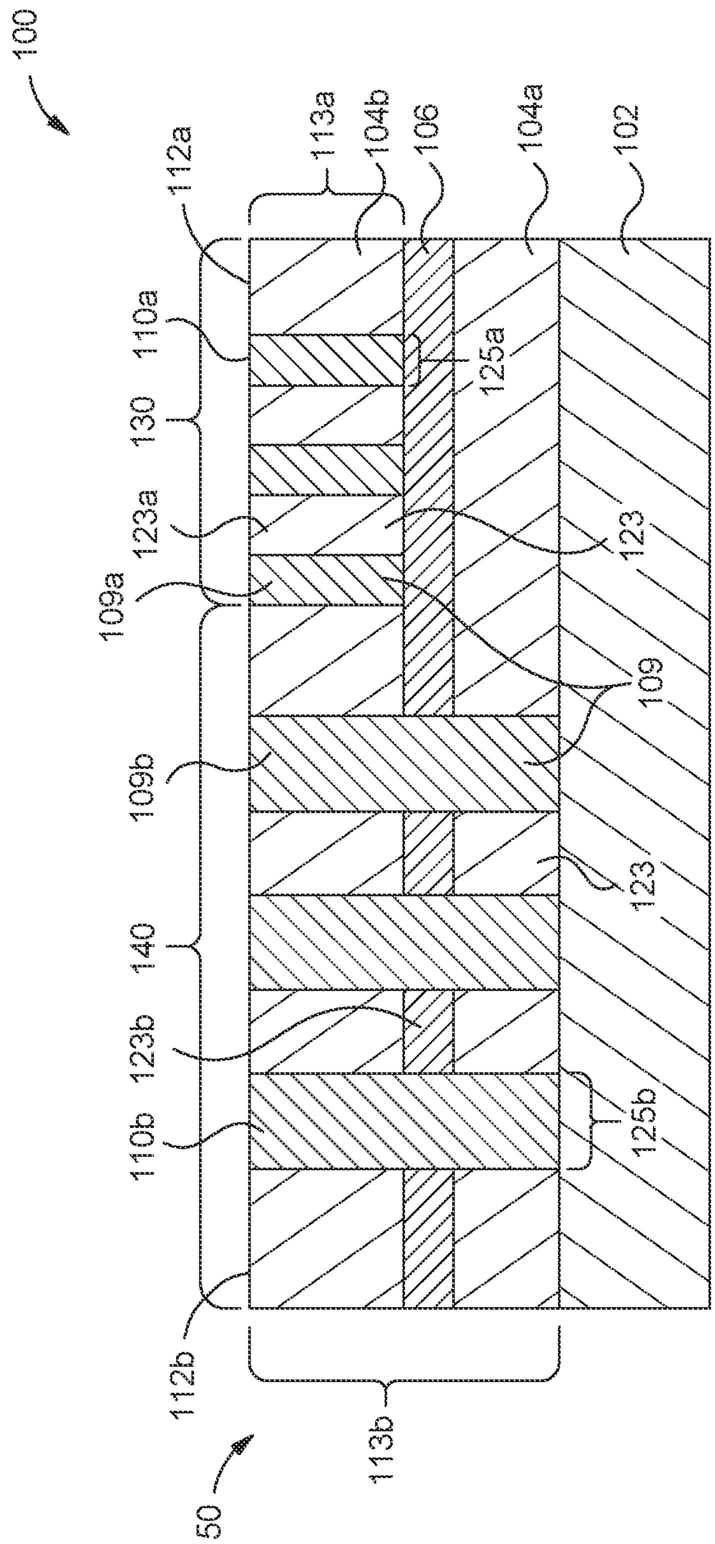
G02B 6/124	(2006.01)
G02B 27/01	(2006.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,025,031	B2 *	7/2018	Tokushima	G02B 6/136
10,139,563	B2 *	11/2018	Baudot	G02B 6/136
10,241,244	B2 *	3/2019	Miller	G02B 5/1857
10,274,678	B1 *	4/2019	Liu	H01L 21/30608
11,002,915	B2 *	5/2021	Kuo	G02B 6/30
11,512,385	B2 *	11/2022	Olson	C23C 14/221
12,050,344	B2 *	7/2024	Messer	G02B 6/1223
2008/0265338	A1 *	10/2008	Yu	H10D 30/024 257/397
2014/0193115	A1 *	7/2014	Popovic	G02B 6/124 385/14
2015/0117817	A1 *	4/2015	Chen	G02B 6/305 359/569
2017/0192170	A1	7/2017	Baudot et al.	
2018/0358777	A1	12/2018	Chen et al.	
2020/0194217	A1	6/2020	Godet et al.	
2020/0257025	A1	8/2020	Tekolste et al.	
2021/0057874	A1	2/2021	Yu	

* cited by examiner



100
112a
113a
104b
106
104a
102
110a
125a
130
123a
123
109a
109
109b
123
140
123b
110b
125b
112b
113b
150

FIG. 1

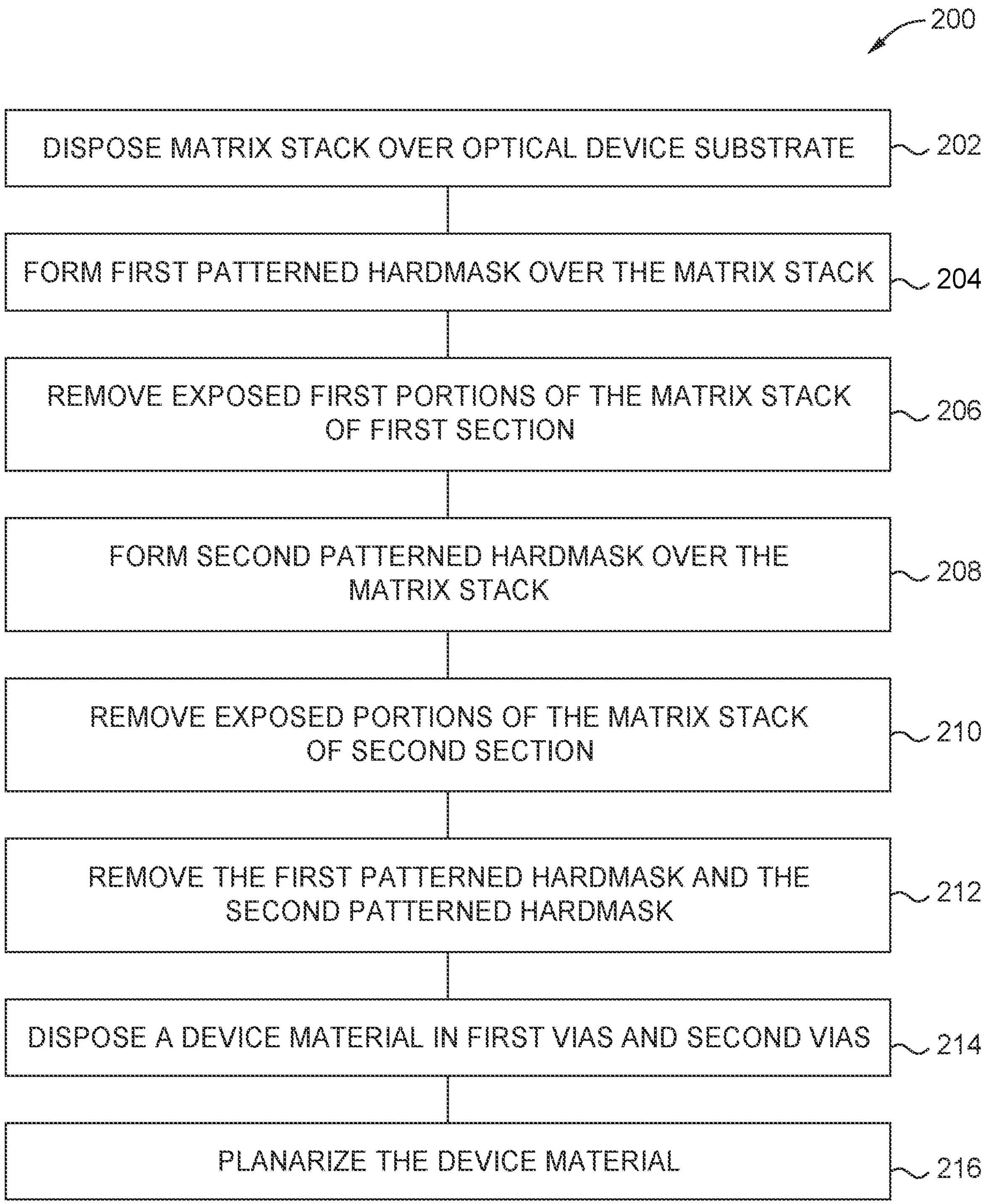

200
DISPOSE MATRIX STACK OVER OPTICAL DEVICE SUBSTRATE
202
FORM FIRST PATTERNED HARDMASK OVER THE MATRIX STACK
204
REMOVE EXPOSED FIRST PORTIONS OF THE MATRIX STACK OF FIRST SECTION
206
FORM SECOND PATTERNED HARDMASK OVER THE MATRIX STACK
208
REMOVE EXPOSED PORTIONS OF THE MATRIX STACK OF SECOND SECTION
210
REMOVE THE FIRST PATTERNED HARDMASK AND THE SECOND PATTERNED HARDMASK
212
DISPOSE A DEVICE MATERIAL IN FIRST VIAS AND SECOND VIAS
214
PLANARIZE THE DEVICE MATERIAL
216

FIG. 2

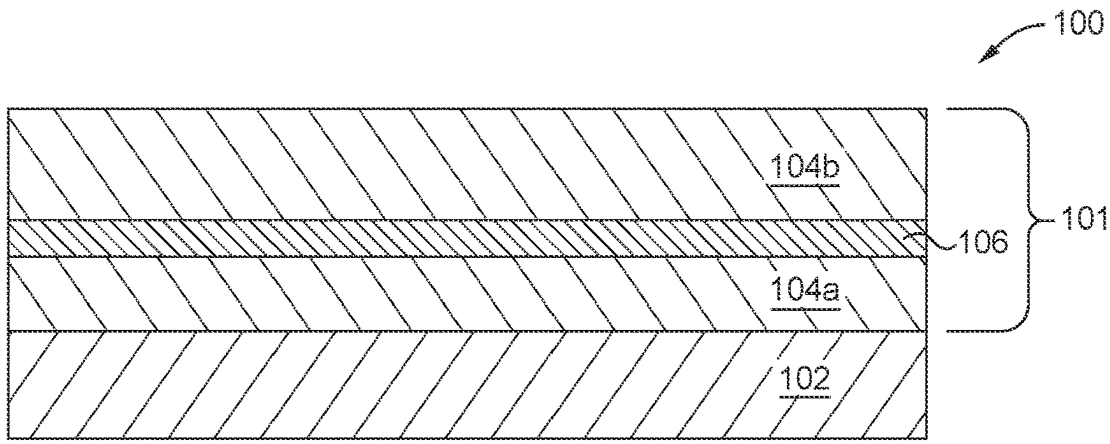


FIG. 3A

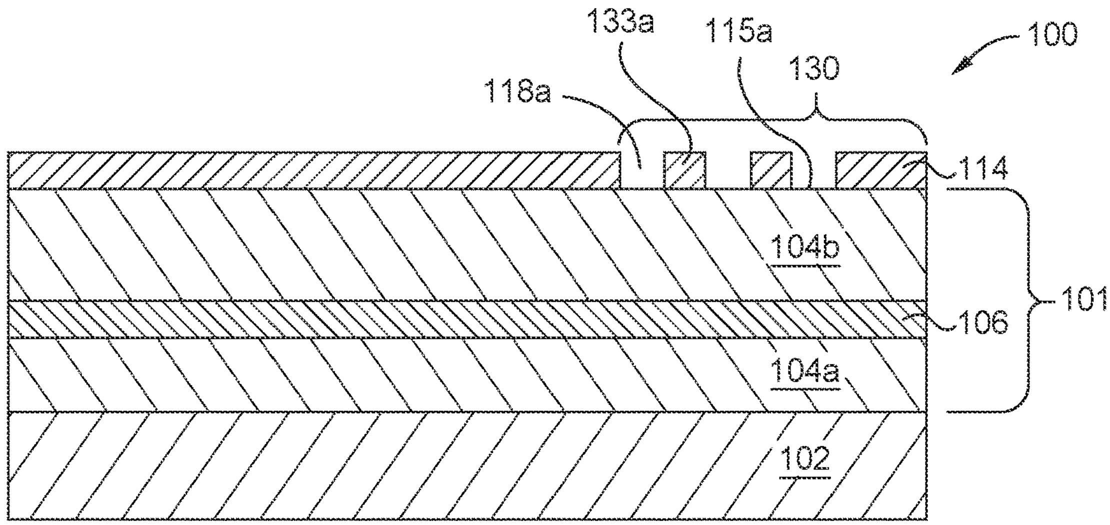


FIG. 3B

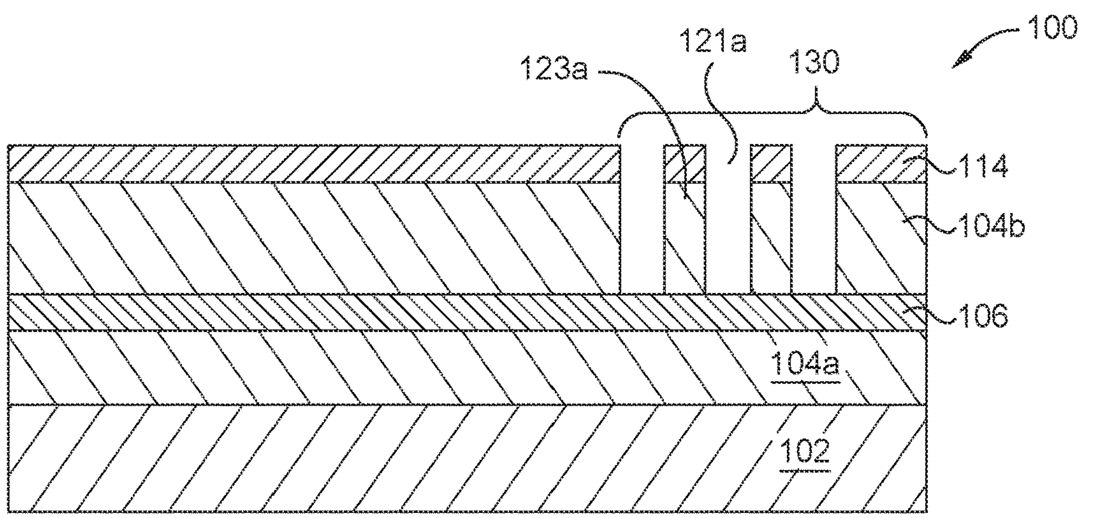


FIG. 3C

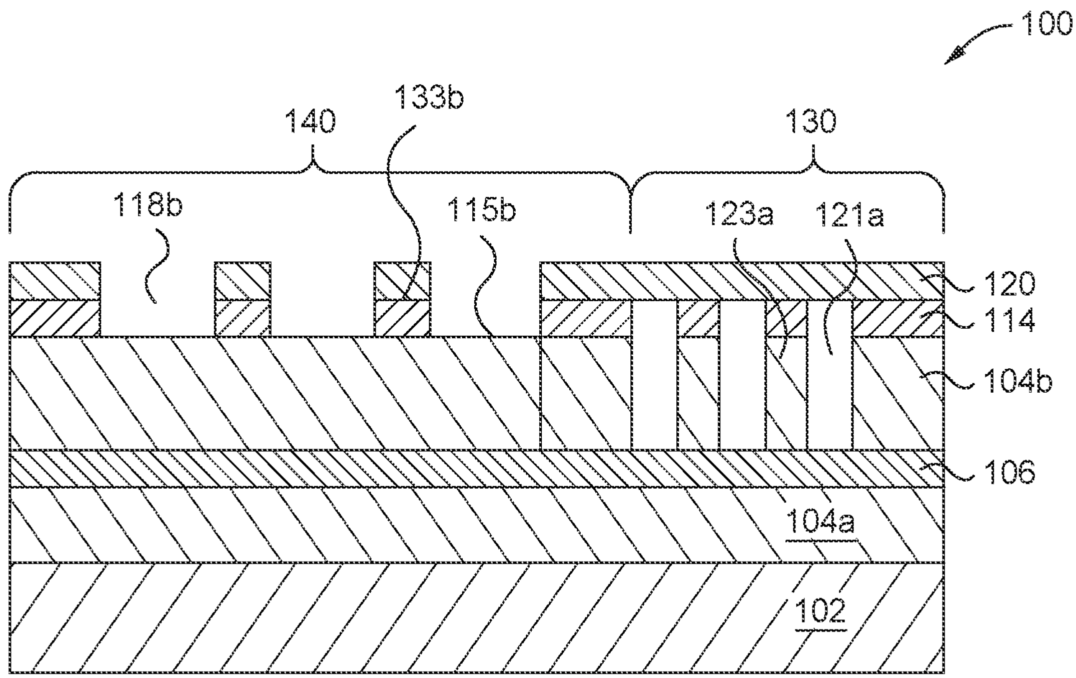

100
140
133b
130
118b
115b
123a
121a
120
114
104b
106
104a
102

FIG. 3D

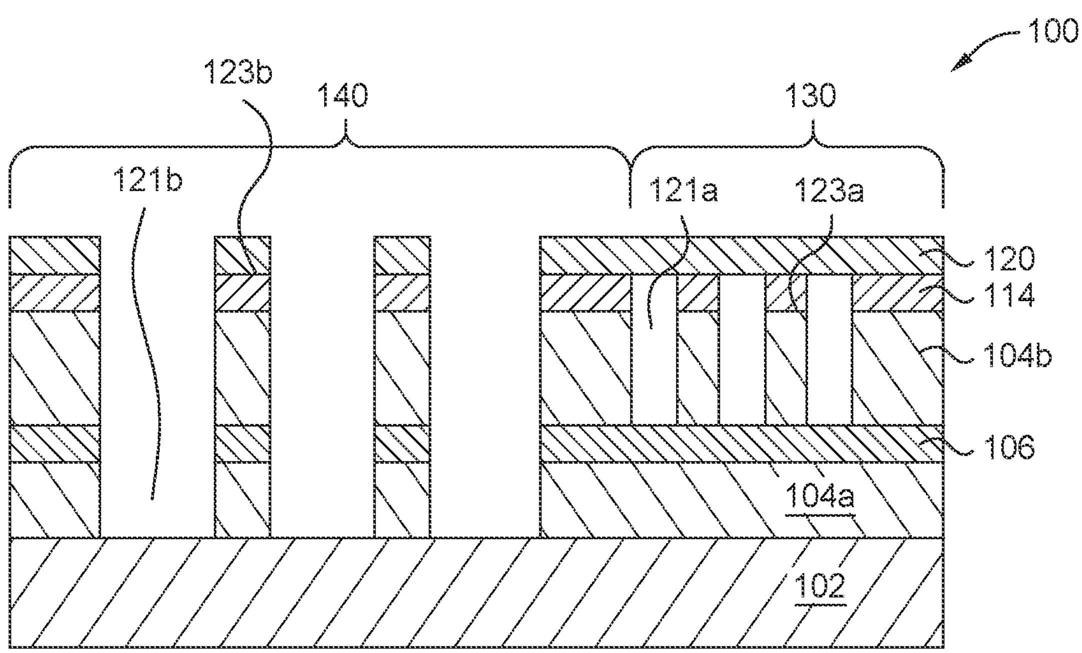

100
123b
140
130
121b
121a
123a
120
114
104b
106
104a
102

FIG. 3E

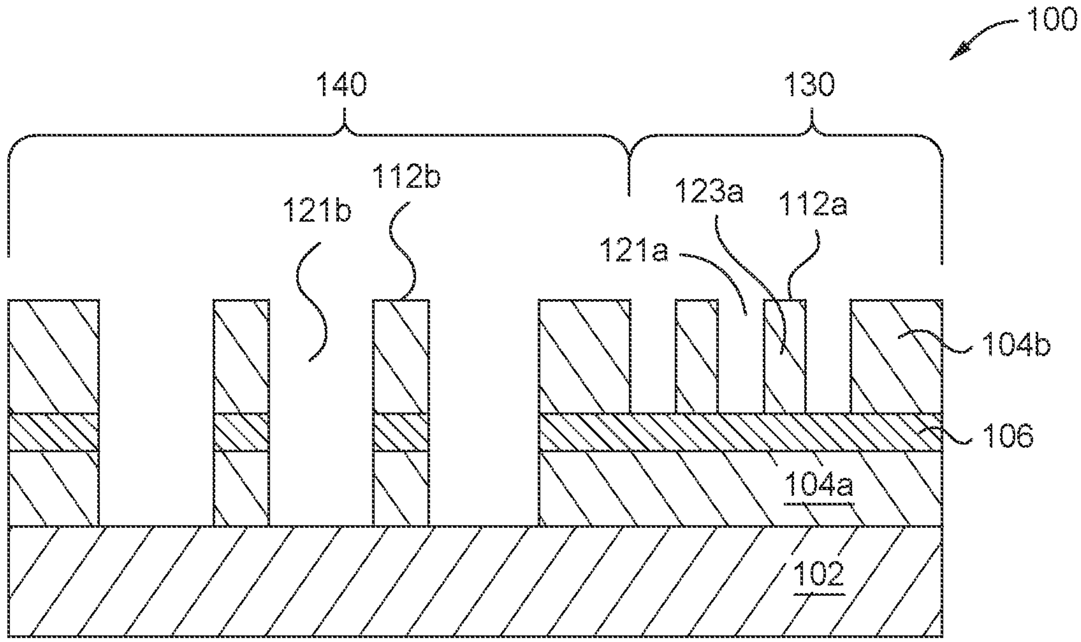

100
140
130
121b
112b
123a
112a
121a
104b
106
104a
102

FIG. 3F

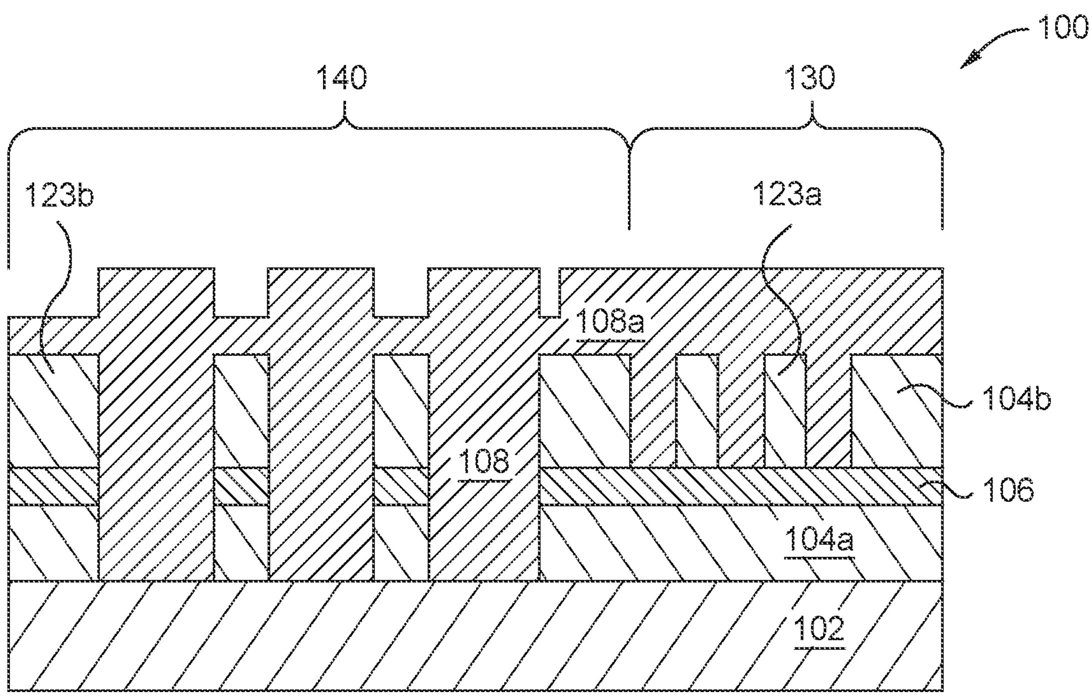

100
140
130
123b
123a
108a
104b
108
106
104a
102

FIG. 3G

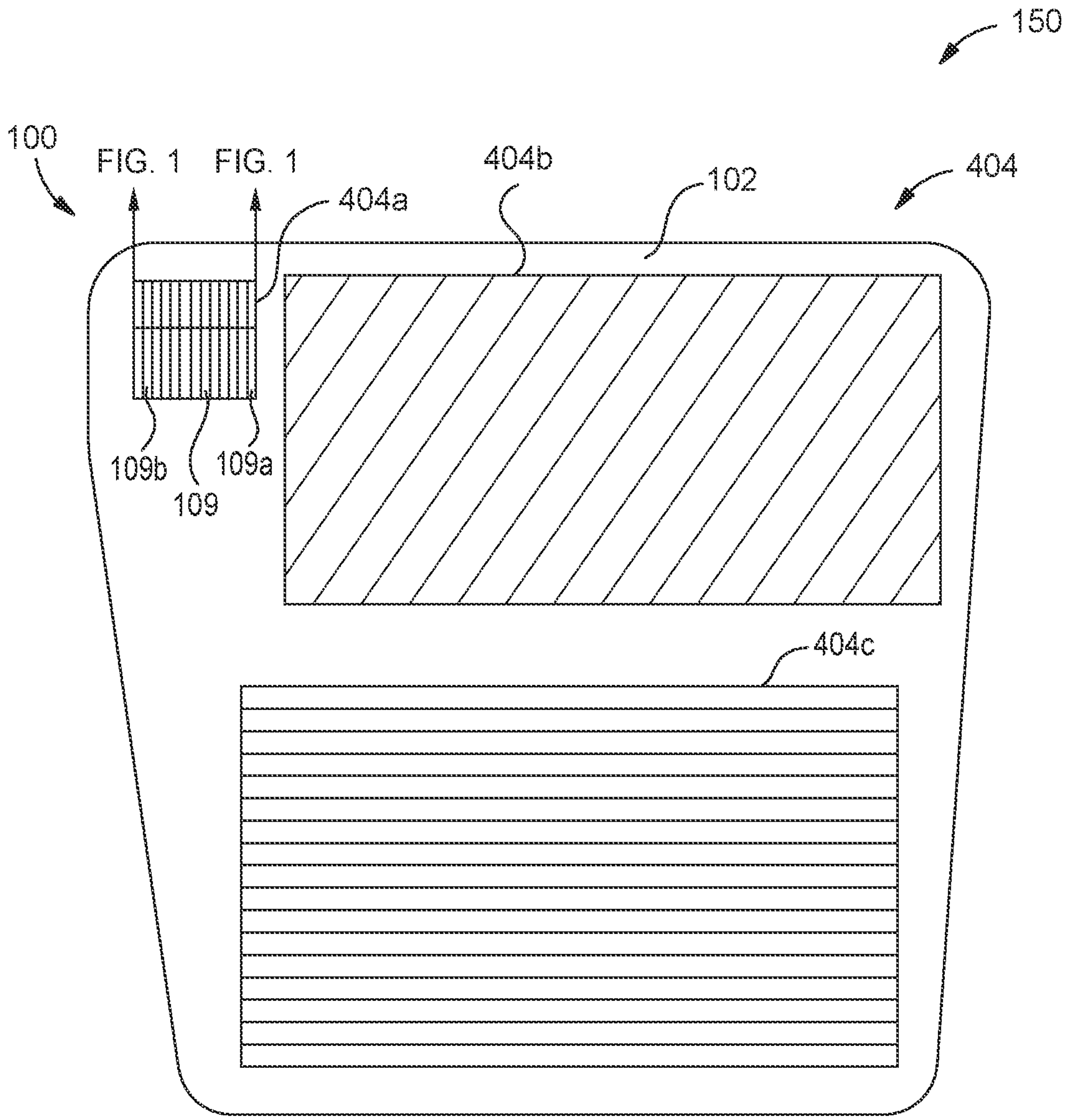


FIG. 4A

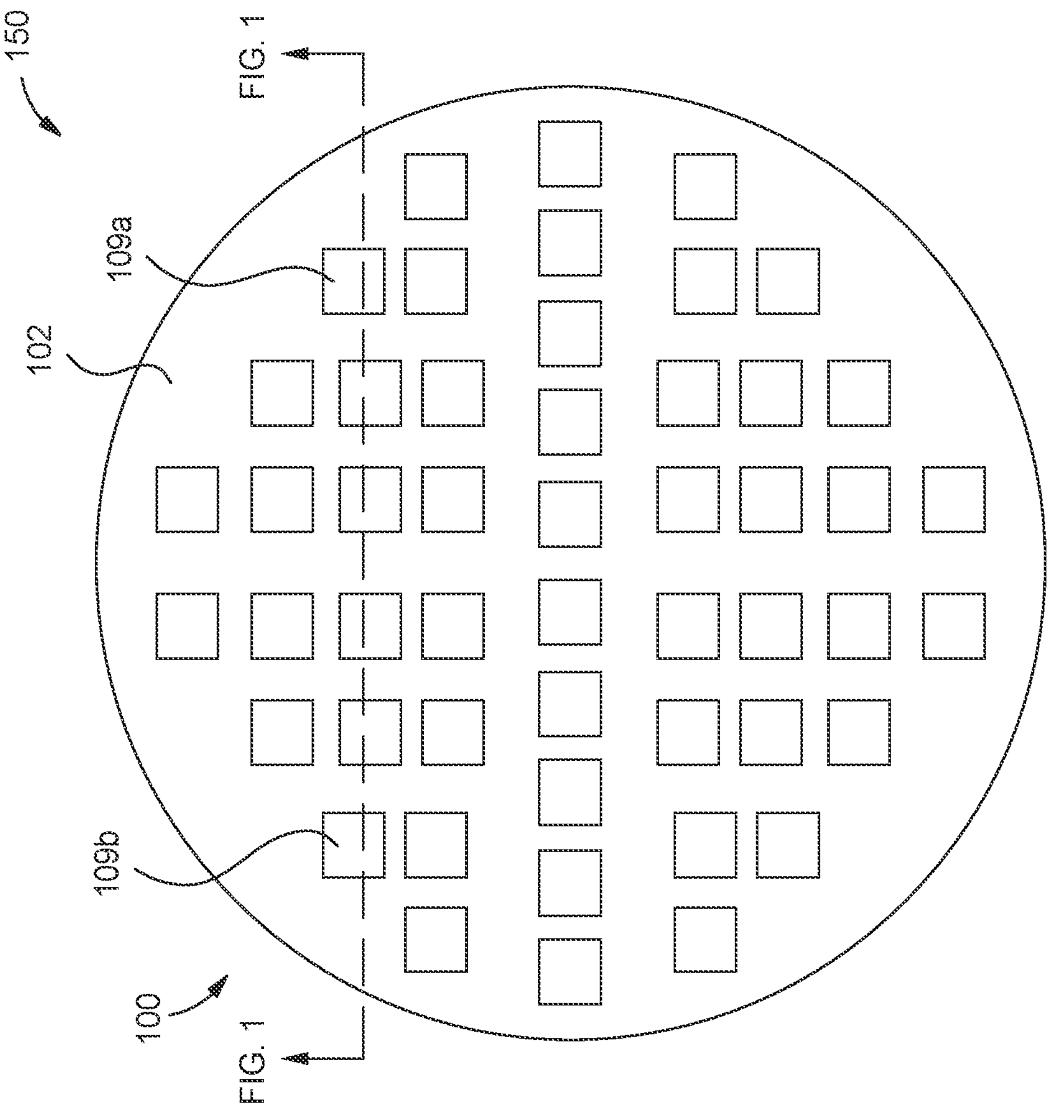


FIG. 4B

METHOD FOR FORMING MULTI-DEPTH OPTICAL DEVICES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims benefit of U.S. Provisional Patent Application No. 63/265,910, filed Dec. 22, 2021, which is herein incorporated by reference.

BACKGROUND

Field

Embodiments of the present disclosure generally relate to optical devices for augmented, virtual, and mixed reality. More specifically, embodiments described herein relate to an optical device and a method of forming an optical device having multi-depth optical device structures with a refractive index greater than or equal to 2.0.

Description of the Related Art

An optical device may be used to manipulate the propagation of light using structures of the optical device formed on a substrate. The optical device includes an arrangement of structures with in-plane dimensions smaller than half a design wavelength of light. The structures have sub-micron critical dimensions, e.g., nanosized dimensions, to alter light propagation by manipulating photons in order to induce localized phase discontinuities (i.e., abrupt changes of phase over a distance smaller than the wavelength of light). In addition to having sub-micron critical dimensions, it is desirable for different sections of the optical device to have structures with different heights. It is also desirable to have a high contrast ratio between the refractive index of the structures and the refractive index of the area between the structures to improve the performance of the optical device.

However, forming an optical device having sub-micron critical dimensions from a high refractive index material, i.e. a material with a refractive index greater than about 2.0, may be challenging. For example, etching a layer of high refractive index material to form structures of differing heights may result in non-uniform critical dimensions. Accordingly, what is needed in the art is an optical device and a method of forming an optical device having multi-depth optical device structures with the refractive index greater than or equal to 2.0.

SUMMARY

In one embodiment, a method is provided. The method includes disposing a matrix stack over a substrate, the substrate having at least a first section and a second section, forming a first patterned hardmask over the matrix stack, the first patterned hardmask having first hardmask structures over the first section, removing exposed first portions of the matrix stack of the first section to form first matrix stack structures, the first matrix stack structures defining first vias, forming a second patterned hardmask over the matrix stack, the second patterned hardmask having second hardmask structures over the second section, removing exposed second portions of the matrix stack of the second section to form second matrix stack structures, the second matrix stack structures defining second vias, removing the first patterned hardmask and second patterned hardmask, and forming first optical device structures over the first section and second optical device structures over the second section, the forming the first optical device structures and the second optical device structures comprising disposing a device material in the first vias and the second vias.

In another embodiment, a method is provided. The method includes disposing a matrix stack of a first matrix layer, an etch-stop layer, and a second matrix layer over a substrate, the substrate having at least a first section and a second section, forming a first patterned hardmask over the matrix stack, the first patterned hardmask having first hardmask structures over the first section, removing exposed first portions of the second matrix layer of the first section to form first matrix stack structures, the first matrix stack structures defining first vias, forming a second patterned hardmask over the matrix stack, the second patterned hardmask having second hardmask structures over the second section, removing exposed second portions of the second matrix layer, the etch-stop layer, and the first matrix layer of the second section to form second matrix stack structures, the second matrix stack structures defining second vias, and forming first optical device structures over the first section and second optical device structures over the second section, the forming the first optical device structures and the second optical device structures comprising disposing a device material in the first vias and the second vias, wherein the first optical device structures have a first depth from the etch-stop layer to a first upper surface of the first optical device structures and the second optical device structures having a second depth from the substrate to a second upper surface of the second optical device structures.

In yet another embodiment, an optical device is provided. The optical device includes a first section of first optical device structures, wherein first matrix stack structures defining the first optical device structures, the first matrix stack structures include a second matrix layer disposed over a first matrix layer, and the first optical device structures have a first depth from the first matrix layer to a first upper surface of the first optical device structures. The optical device also includes a second section of second optical device structures, wherein second matrix stack structures defining the second optical device structures, the second matrix stack structures include the first matrix layer and the second matrix layer disposed over the first matrix layer, and the second optical device structures have a second depth from an optical device substrate to a second upper surface of the second optical device structures.

BRIEF DESCRIPTION OF THE DRAWINGS

So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, and may admit to other equally effective embodiments.

FIG. **1** is a schematic, cross-sectional view of a portion of an optical device according to one or more embodiments.

FIG. **2** is a flow diagram of a method of forming an optical device according to one or more embodiments.

FIGS. **3**A-**3**G are schematic, cross-sectional views of a portion of an optical device substrate according to one or more embodiments.

FIG. **4**A is a perspective, frontal view of an optical device according to embodiments described herein.

FIG. 4B is schematic, top view of an optical device according to embodiments described herein.

To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

Embodiments of the present disclosure generally relate to optical devices for augmented, virtual, and mixed reality. More specifically, embodiments described herein relate to an optical device and a method of forming an optical device having multi-depth optical device structures with a refractive index greater than or equal to 2.0.

FIG. **1** is a schematic, cross-sectional view of a portion **100** of an optical device substrate **102** according to one or more embodiments. In one embodiment, which can be combined with other embodiments described herein, the optical device **150** is a flat optical device, such as a metasurface. In another embodiment, which can be combined with other embodiments described herein, the optical device **150** is a waveguide combiner, such as an augmented waveguide combiner. The portion **100** may correspond to a portion of a flat optical device or a portion of a waveguide combiner. In one example, the portion **100** corresponds to a grating of a waveguide combiner, such as one or more of an input coupling grating, an intermediate grating, or an output coupling grating. The portion **100** may correspond to a cross-section as depicted in FIGS. **4**A and **4**B.

The optical device **150** includes a plurality of optical device structures **109** disposed over an optical device substrate **102**. A first section **130** of the portion **100** of the optical device substrate **102** includes a plurality of first optical device structures **109***a*. A second section **140** of the portion **100** of the optical device substrate **102** includes a plurality of second optical device structures **109***b*. While the portion **100** of the optical device substrate **102** depicts the first section **130** and the second section **140**, the optical device **150** may include additional sections of optical device structures **109**.

The plurality of optical device structures **109** have a depth and a critical dimension. More specifically, the first optical device structures **109***a* have a depth **113***a* and a critical dimension **125***a*, and the second optical device structures **109***b* have a depth **113***b* and a critical dimension **125***b*. The critical dimension **125***a* corresponds to a width or a diameter of the first optical device structures **109***a* and the critical dimension **125***b* correspond to the width or the diameter of the second optical device structures **109***b*. In some embodiments, the critical dimension **125***a* and the critical dimension **125***b* are the same. In other embodiments, the critical dimension **125***a* and the critical dimension **125***b* are different. The critical dimension **125***a* and the critical dimension **125***b* are less than 1 micrometer (μm). In one embodiment, which may be combined with other embodiments described herein, the critical dimension **125***a* and the critical dimension **125***b* are about 100 nanometers (nm) to about 1000 nm. The depth **113***a* is the distance from an optional etch-stop layer **106** to a first upper surface **110***a* of the first optical device structures **109***a*. The depth **113***b* is the distance from the substrate to a second upper surface **110***b* of the second optical device structures **109***b*. The depth **113***a* and the depth **113***b* are different. The etch-stop layer **106** is an optional layer. In embodiments without an optional etch-stop layer **106**, the depth **113***a* corresponds to the thickness of the second matrix layer **104***b*.

The first optical device structures **109***a* may have a first aspect ratio defined by a ratio of the depth **113***a* to the critical dimension **125***a*. The second optical device structures **109***b* may have a second aspect ratio defined by a ratio of the depth **113***b* to the critical dimension **125***b*. The first aspect ratio and the second aspect ratio are about 1:1.5 to about 1:20, such as about 1:10 to about 1:15. The first aspect ratio and the second aspect ratio may be different from each other.

The plurality of optical device structures **109** have a device refractive index. The device refractive index is greater than about 2.0. The plurality of optical device structures **109** include, but are not limited to, one or more of silicon (e.g. amorphous silicon), silicon carbide (SiC), silicon oxycarbide (SiOC), titanium dioxide (TiO_2), silicon dioxide (SiO_2), vanadium (IV) oxide (VOx), aluminum oxide (Al_2O_3), aluminum-doped zinc oxide (AZO), indium tin oxide (ITO), tin dioxide (SnO_2), zinc oxide (ZnO), tantalum pentoxide (Ta_2O_5), silicon nitride (Si_3N_4), zirconium dioxide (ZrO_2), niobium oxide (Nb_2O_5), cadmium stannate (Cd_2SnO_4), or silicon carbon-nitride (SiCN) containing materials.

The optical device **150** includes a plurality of matrix stack structures **123**. Adjacent matrix stack structures **123** include an optical device structure **109** disposed therebetween. Each of the matrix stack structures **123** include at least a first matrix layer **104***a* disposed over the optical device substrate **102**, an optional etch-stop layer **106** disposed over the first matrix layer **104***a*, and a second matrix layer **104***b* disposed over the optional etch-stop layer **106**. The first section **130** of the portion **100** of the optical device substrate **102** includes a plurality of first matrix stack structures **123***a* and the second section **140** of the portion **100** of the optical device substrate **102** includes a plurality of second matrix stack structures **123***b*. Adjacent first optical device structures **109***a* include a first matrix stack structure **123***a* disposed therebetween. Adjacent second optical device structures **109***b* include a second matrix stack structure **123***b* disposed therebetween.

Each of the plurality of first optical device structures **109***a* include a bottom surface contacting the optional etch-stop layer **106** and side surfaces contacting adjacent first matrix stack structures **123***a*. The first upper surface **110***a* of the first optical device structures **109***a* aligns with a first upper surface **112***a* of the first matrix stack structures **123***a*. Each of the plurality of second optical device structures **109***b* include a bottom surface contacting the optical device substrate **102** and side surfaces contacting adjacent second matrix stack structures **123***b*. The second upper surface **110***b* of the second optical device structures **109***b* aligns with a second upper surface **112***b* of the second matrix stack structures **123***b*. The depth **113***a* is about 100 nm to about 750 nm corresponding to a thickness of the second matrix layer **104***b*. In embodiments including the etch-stop layer **106**, the depth **113***b* is about 200 nm to about 1550 nm corresponding to a total thickness of the first matrix layer **104***a*, the optional etch-stop layer **106**, and the second matrix layer **104***b*. In embodiments without the etch-stop layer **106**, the depth **113***b* is about 200 nm to about 1500 nm corresponding to a total thickness of the first matrix layer **104***a* and the second matrix layer **104***b*. The thickness of the first matrix layer **104***a* is about 100 nm to about 750 nm. The thickness of the optional etch-stop layer **106** is less than 50 nm.

The matrix stack structures **123** have a matrix stack refractive index corresponding to an average of the refractive index of the materials of the first matrix layer **104***a*, the optional etch-stop layer **106**, and the second matrix layer **104***b*. The matrix stack refractive index is less than 2.0, such as less than about 1.5. The contrast ratio between the device refractive index and the matrix stack refractive index is from about 1.7:1.45 to about 3.7:1.45. The optional etch-stop layer **106** may include a same material as or a different material than the optical device substrate **102**. The optional etch-stop layer **106** includes a same material as or a different material than the first matrix layer **104***a*. The optional etch-stop layer **106** includes a same material as or a different material than the first matrix layer **104***a* and the second matrix layer **104***b*. The second matrix layer **104***b* includes a same material as or a different material than the first matrix layer **104***a*.

The first matrix layer **104***a* and the second matrix layer **104***b* include, but are not limited to, silicon dioxide (SiO_2). The optional etch-stop layer **106** includes, but is not limited to, one or more of silicon (e.g. amorphous silicon), silicon carbide (SiC), silicon oxycarbide (SiOC), titanium dioxide (TiO_2), silicon dioxide (SiO_2), vanadium (IV) oxide (VOx), aluminum oxide (Al_2O_3), aluminum-doped zinc oxide (AZO), indium tin oxide (ITO), tin dioxide (SnO_2), zinc oxide (ZnO), tantalum pentoxide (Ta_2O_5), silicon nitride (Si_3N_4), zirconium dioxide (ZrO_2), niobium oxide (Nb_2O_5), cadmium stannate (Cd_2SnO_4), or silicon carbon-nitride (SiCN) containing materials.

The optical device substrate **102** may be any suitable material, provided that the optical device substrate **102** can adequately transmit light in a desired wavelength or wavelength range. Substrate selection may include substrates of any suitable material, including, but not limited to, amorphous dielectrics, non-amorphous dielectrics, crystalline dielectrics, silicon oxide, polymers, and combinations thereof. In some embodiments, which may be combined with other embodiments described herein, the optical device substrate **102** includes a transparent material. In one embodiment, which can be combined with other embodiments described herein, the optical device substrate **102** includes, but is not limited to, one or more of silicon (Si), silicon dioxide (SiO_2), silicon carbide (SiC), germanium (Ge), silicon germanium (SiGe), indium phosphide (InP), gallium arsenide (GaAs), gallium nitride (GaN), fused silica, quartz, or sapphire. In another embodiment, which can be combined with other embodiments described herein, the optical device substrate **102** includes high-index transparent materials, such as high-refractive-index (high RI) glass.

FIG. **2** is a flow diagram of a method **200** of forming an optical device **150**. FIGS. **3**A-**3**G are schematic, cross-sectional views of a portion **100** of an optical device substrate **102**.

At operation **202**, as shown in FIG. **3**A, a matrix stack **101** is disposed over the optical device substrate **102**. The matrix stack **101** includes the first matrix layer **104***a*, the optional etch-stop layer **106**, and the second matrix layer **104***b*. The first matrix layer **104***a* is disposed between the optical device substrate **102** and the optional etch-stop layer **106**. The second matrix layer **104***b* is disposed over the optional etch-stop layer **106**. The first matrix layer **104***a* has a thickness substantially equal to or a thickness different from the second matrix layer **104***b*.

At operation **204**, as shown in FIG. **3**B, a first patterned hardmask **114** is formed over the matrix stack **101**. The first patterned hardmask **114** includes first hardmask structures **133***a* formed over the first section **130** of the optical device substrate **102**. The first hardmask structures **133***a* form exposed first portions **115***a* of the matrix stack **101** of the first section **130** by defining first contact holes **118***a* of the first patterned hardmask **114**. Forming the first patterned hardmask **114** includes disposing a first hardmask material over the matrix stack **101**, disposing a photoresist over the first hardmask material, patterning the photoresist to expose portions of the first hardmask material, and removing the exposed portions of the first hardmask material to form the first hardmask structures **133***a*. The first hardmask material may include, but is not limited to silicon nitride (Si_3N_4).

At operation **206**, as shown in FIG. **3**C, the exposed first portions **115***a* of the matrix stack **101** of the first section **130** are removed. Removing the exposed first portions **115***a* of the matrix stack **101** forms first matrix stack structures **123***a*. The first matrix stack structures **123***a* define first vias **121***a* such that two adjacent first matrix stack structures **123***a* have a first via **121***a* disposed therebetween. The forming of the first matrix stack structures **123***a* includes removing the second matrix layer **104***b* to expose the optional etch-stop layer **106**. In an optional operation, at least one of the first vias **121***a*, the second vias **121***b*, the first contact holes **118***a*, or the second contact holes **118***b* are shrunk.

At operation **208**, as shown in FIG. **3**D, a second patterned hardmask **120** is formed over the matrix stack **101**. The second patterned hardmask **120** includes second hardmask structures **133***b* formed over the second section **140** of the optical device substrate **102**. The second hardmask structures **133***b* form exposed second portions **115***b* of the matrix stack **101** of the second section **140** by defining second contact holes **118***b* of the second patterned hardmask **120**. Forming the second patterned hardmask **120** includes disposing a second hardmask material over the matrix stack **101**, disposing a photoresist over the second hardmask material, patterning the photoresist to expose portions of the second hardmask material, and removing the exposed portions of the second hardmask material to form the second hardmask structures **133***b*. The second hardmask material may include, but is not limited to silicon nitride (Si_3N_4).

At operation **210**, as shown in FIG. **3**E, the exposed second portions **115***b* of the matrix stack **101** of the second section **140** are removed. Removing the exposed second portions **115***b* forms second matrix stack structures **123***b*. The second matrix stack structures **123***b* define second vias **121***b* such that two adjacent second matrix stack structures **123***b* have a second via **121***b* disposed therebetween. The forming of the second matrix stack structures **123***b* includes removing the second matrix layer **104***b*, the optional etch-stop layer **106**, and the first matrix layer **104***a* to expose the optical device substrate **102**.

At operation **212**, as shown in FIG. **3**F, the first patterned hardmask **114** and the second patterned hardmask **116** are removed. Residual photoresist materials are also removed.

At operation **214**, as shown in FIG. **3**G, a device material **108** is disposed in the first vias **121***a* and second vias **121***b*. In one example, the device material **108** is disposed by Atomic Layer Deposition (ALD). The device material may include one or more of silicon oxycarbide (SiOC), titanium dioxide (TiO_2), silicon dioxide (SiO_2), vanadium (IV) oxide (VOx), aluminum oxide (Al_2O_3), aluminum-doped zinc oxide (AZO), indium tin oxide (ITO), tin dioxide (SnO_2), zinc oxide (ZnO), tantalum pentoxide (Ta_2O_5), silicon nitride (Si_3N_4), zirconium dioxide (ZrO_2), niobium oxide (Nb_2O_5), cadmium stannate (Cd_2SnO_4), or silicon carbon-nitride (SiCN) containing materials.

At operation **216**, as shown in FIG. **1**, the device material **108** is planarized. The device material **108** is planarized to form the first optical device structures **109***a* and the second optical device structures **109***b* such that the first upper surface **110***a* of the first optical device structures **109***a* aligns with the first upper surface **112***a* of the first matrix stack structures **123***a* and the second upper surface **110***b* of the second optical device structures **109***b* aligns with the second upper surface **112***b* of the second matrix stack structures **123***b*. In one example, the excess portion **108***a* of the device material **108** may be removed using chemical-mechanical planarization (CMP).

FIG. **4**A is a perspective, frontal view of an optical device **150** of a first configuration. FIG. **4**B is a schematic, top view of an optical device **150** of a first configuration. It is to be understood that the optical devices **150** of the first and second configurations described are exemplary optical devices. In one embodiment, which can be combined with other embodiments described herein, the optical device **150** of the first configuration is a waveguide combiner, such as an augmented reality waveguide combiner. In another embodiment, which can be combined with other embodiments described herein, the optical device **150** of the second configuration is a flat optical device, such as a metasurface. The optical devices **150** include the plurality of optical device structures **109** disposed the optical device substrate **102**. The plurality of optical device structures **109** include the plurality of first optical device structures **109***a* and plurality of second optical device structures **109***b*. In one embodiment, which can be combined with other embodiments described herein, regions of the optical device structures **109** correspond to one or more gratings **404**, such as a first grating **404***a*, a second grating **404***b*, and a third grating **404***c*. In one embodiment, which can combined with other embodiments described herein, the optical devices **150** is a waveguide combiner that includes at least the first grating **404***a* corresponding to an input coupling grating and the third grating **404***c* corresponding to an output coupling grating. The waveguide combiner according to the embodiment, which can be combined with other embodiments described herein, includes the second grating **404***b* corresponding to an intermediate grating. While FIG. **4**B depicts the optical device structures **109** as having square or rectangular shaped cross-sections, the cross-sections of the optical device structures **109** may have other shapes including, but not limited to, circular, triangular, elliptical, regular polygonal, irregular polygonal, and/or irregular shaped cross-sections. In some embodiments, which can be combined with other embodiments described herein, the cross-sections of the optical device structures **109** on a single optical device **150** are different.

In summation, an optical device and a method of forming an optical device having multi-depth optical device structures with a refractive index greater than or equal to 2.0 are provided. The optical devices and method include forming first matrix stack structures and second matrix stack structures. Adjacent first matrix stack structures form first vias having a first depth and adjacent second matrix stack structures form second vias having a second depth. The first depth of the first vias being different than the second depth of the second vias provides from the formation of submicron, multi-depth optical device structures when the vias are backfilled with a device material.

While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

What is claimed is:

1. A method, comprising:

disposing a matrix stack over a substrate, the substrate having at least a first section and a second section;

forming a first patterned hardmask over the matrix stack, the first patterned hardmask having first hardmask structures over the first section;

removing exposed first portions of the matrix stack of the first section to form first matrix stack structures, the first matrix stack structures defining first vias;

forming a second patterned hardmask over the matrix stack, the second patterned hardmask having second hardmask structures over the second section;

removing exposed second portions of the matrix stack of the second section to form second matrix stack structures, the second matrix stack structures defining second vias;

removing the first patterned hardmask and second patterned hardmask; and

forming first optical device structures over the first section and second optical device structures over the second section, the forming the first optical device structures and the second optical device structures comprising disposing a device material in the first vias and the second vias, wherein:

the first matrix stack structures define the first optical device structures;

the first matrix stack structures include a second matrix layer disposed over a first matrix layer;

the first optical device structures have a first depth from the first matrix layer to a first upper surface of the first optical device structures;

a bottom surface of the first optical device structures contact an etch stop layer;

the second matrix stack structures define the second optical device structures;

the second matrix stack structures include the first matrix layer and the second matrix layer disposed over the first matrix layer;

the second optical device structures have a second depth from an optical device substrate to a second upper surface of the second optical device structures; and

a bottom surface of the second optical device structures contact a top surface of a substrate.

2. The method of claim **1**, wherein the device material has a device refractive index greater than or equal to 2.0.

3. The method of claim **2**, wherein the first matrix stack structures and the second matrix stack structures have a matrix stack refractive index less than 2.0.

4. The method of claim **3**, wherein a contrast ratio between the device refractive index and the matrix stack refractive index is about 1.7:1.45 to about 3.7:1.45.

5. The method of claim **1**, wherein the matrix stack comprises a first matrix layer, an etch-stop layer, and a second matrix layer.

6. The method of claim **5**, wherein:

the removing exposed first portions of the matrix stack of the first section includes removing the second matrix layer to expose the etch-stop layer; and

the removing exposed second portions of the matrix stack of the second section includes removing the first matrix layer, the etch-stop layer, and the second matrix layer to expose the substrate.

7. The method of claim 6, wherein:

the first optical device structures have a first depth from the etch-stop layer to a first upper surface of the first optical device structures; and

the second optical device structures have a second depth from the substrate to a second upper surface of the second optical device structures.

8. The method of claim 1, wherein:

adjacent first hardmask structures define first contact holes; and

adjacent second hardmask structures define second contact holes.

9. The method of claim 8, further comprising shrinking at least one of the first vias.

10. The method of claim 1, further comprising planarizing the device material.

11. The method of claim 1, wherein the device material comprises one or more of silicon oxycarbide (SiOC), titanium dioxide (TiO_2), silicon dioxide (SiO_2), vanadium (IV) oxide (VOX), aluminum oxide (Al_2O_3), aluminum-doped zinc oxide (AZO), indium tin oxide (ITO), tin dioxide (SnO_2), zinc oxide (ZnO), tantalum pentoxide (Ta_2O_5), silicon nitride (Si_3N_4), zirconium dioxide (ZrO_2), niobium oxide (Nb_2O_5), cadmium stannate (Cd_2SnO_4), or silicon carbon-nitride (SiCN) containing materials.

12. A method, comprising:

disposing a matrix stack of a first matrix layer, an etch-stop layer, and a second matrix layer over a substrate, the substrate having at least a first section and a second section;

forming a first patterned hardmask over the matrix stack, the first patterned hardmask having first hardmask structures over the first section;

removing exposed first portions of the second matrix layer of the first section to form first matrix stack structures, the first matrix stack structures defining first vias;

forming a second patterned hardmask over the matrix stack, the second patterned hardmask having second hardmask structures over the second section;

removing exposed second portions of the second matrix layer, the etch-stop layer, and the first matrix layer of the second section to form second matrix stack structures, the second matrix stack structures defining second vias; and

forming first optical device structures over the first section and second optical device structures over the second section, the forming the first optical device structures and the second optical device structures comprising disposing a device material in the first vias and the second vias, wherein:

the first optical device structures have a first depth from the etch-stop layer to a first upper surface of the first optical device structures; and

the second optical device structures have a second depth from the substrate to a second upper surface of the second optical device structures, wherein:

the first matrix stack structures define the first optical device structures;

the first matrix stack structures include a second matrix layer disposed over a first matrix layer;

the first optical device structures have a first depth from the first matrix layer to a first upper surface of the first optical device structures;

a bottom surface of the first optical device structures contact an etch stop layer;

the second matrix stack structures define the second optical device structures;

the second matrix stack structures include the first matrix layer and the second matrix layer disposed over the first matrix layer;

the second optical device structures have a second depth from an optical device substrate to a second upper surface of the second optical device structures; and

a bottom surface of the second optical device structures contact a top surface of a substrate.

13. The method of claim 12, further comprising shrinking at least one of the first vias.

14. An optical device, comprising:

a first section of first optical device structures, wherein:

first matrix stack structures define the first optical device structures;

the first matrix stack structures include a second matrix layer disposed over a first matrix layer; and

the first optical device structures have a first depth from the first matrix layer to a first upper surface of the first optical device structures;

a bottom surface of the first optical device structures contact an etch stop layer; and

a second section of second optical device structures, wherein:

second matrix stack structures define the second optical device structures;

the second matrix stack structures include the first matrix layer and the second matrix layer disposed over the first matrix layer;

the second optical device structures have a second depth from an optical device substrate to a second upper surface of the second optical device structures; and

a bottom surface of the second optical device structures contact a top surface of a substrate.

15. The optical device of claim 14, wherein the first optical device structures and the second optical device structures have a device refractive index greater than or equal to 2.0.

16. The optical device of claim 15, wherein the first matrix stack structures and the second matrix stack structures have a matrix stack refractive index less than 2.0.

17. The optical device of claim 16, wherein a contrast ratio between the device refractive index and the matrix stack refractive index is about 1.7:1.45 to about 3.7:1.45.

18. The optical device of claim 14, wherein the second matrix layer is disposed on an etch-stop layer, and wherein the second matrix stack structures further include the etch-stop layer.

19. The optical device of claim 15, wherein the first optical device structures and the second optical device structures comprise one or more of silicon oxycarbide (SiOC), titanium dioxide (TiO_2), silicon dioxide (SiO_2), vanadium (IV) oxide (VOX), aluminum oxide (Al_2O_3), aluminum-doped zinc oxide (AZO), indium tin oxide (ITO), tin dioxide (SnO_2), zinc oxide (ZnO), tantalum pentoxide (Ta_2O_5), silicon nitride (Si_3N_4), zirconium dioxide (ZrO_2), niobium oxide (Nb_2O_5), cadmium stannate (Cd_2SnO_4), or silicon carbon-nitride (SiCN) containing materials.

20. The optical device of claim 14, wherein the first matrix layer is about 100 to about 750 nanometers (nm) thick and the second matrix layer is about 100 to about 750 nm thick.

* * * * *